pfi

Python for Fixed Income

COURSE 10 · IMPLEMENTING RAG

Case Study: The First RAG Pipeline

WEEK 1 · DAY 5

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Today: the first RAG pipeline

The parts, joined

Chunks	1,158 at 512/64 (day 3)
Index	MiniLM, FAISS exact (day 4)
Retrieval	dense, k = 3
Questions	the five arc questions
Answers	12 saved; replayed, no key

Today

1. Every setting in one record
2. The retriever
3. The prompt
4. One answer
5. The local route, on a T4
6. Case study: fifteen answers, graded
7. What the pipeline shows, and where it fails

Every answer cites a chunk or says NOT IN CORPUS; every count says which rule it uses.

How the pipeline works

The problem	Day 1's answers came from training, with nothing to check them against
Retrieve	Embed the question with the chunks' model, score every chunk, keep the k best with their citations
Prompt and send	System prompt, the citation rule, the labelled chunks, the question last; one request at temperature 0
Check and grade	Read the citations back: each must name a chunk that was sent; the grader asks whether the key is in the text
You control	Chunker, embedder, index, k, and the request's settings, in one frozen record; out: a cited answer or NOT IN CORPUS

Every setting in one record

RagSettings(index="faiss").settings_line(): computed, never typed

```
chunker=recursive 512/64, embedder=minilm@1110a24, index=faiss exact, retrieve=dense k=3, rerank=none, expand=0, model=gpt-4.1-mini, temperature=0, top_p=1, max_tokens=300
```

6579143b53aa…

settings_hash(), k = 3

9b08e87b0ea7…

settings_hash(), k = 5

frozen

a new setting is a new record

```
FrozenInstanceError: cannot assign to field 'k'
```

A result means nothing unless every setting is written down beside it. The hash names a row of the log.

Every field of RagSettings

13 fields from dataclasses.fields; today's record changes 1 from the default

field	default	today	changed by
size	512	512	day 8: 1,024 in the grid's chunk1024 settings
overlap	64	64	day 8: 128 with size 1,024
embedder	'minilm'	'minilm'	day 4 compared three; the course keeps minilm
index	**'numpy'**	**'faiss'**	**today: faiss (numpy on day 4)**
retrieve	'dense'	'dense'	day 6: bm25 and hybrid
k	3	3	exercise 1: 1; day 8: 2 and 5
candidates	20	20	day 7: the pool a reranker rescores
rerank	'none'	'none'	day 7: cross-encoder
expand	0	0	day 8: one neighbour on each side
model	'gpt-4.1-mini'	'gpt-4.1-mini'	never: gpt-4.1-mini on every recorded request
temperature	0.0	0.0	day 8's request-only settings
top_p	1.0	1.0	day 8's request-only settings; sent only when it is not 1
max_tokens	300	300	day 8's request-only settings

The retriever: arc02, credit facility outstanding

retrieve(question): three chunks, best first; the key ($160.7 million) is on page 37

```
rank 1: boyd10k-7-036-0413, Item 7, page 36, score 0.6802  [boyd10k; Item 7; p. 36; c0413]
rank 2: boyd10k-1A-019-0225, Item 1A, page 19, score 0.6375  [boyd10k; Item 1A; p. 19; c0225]
rank 3: boyd10k-8-082-0917, Item 8, page 82, score 0.6273  [boyd10k; Item 8; p. 82; c0917]
```

The top chunk, quoted as published: holds 160.7: True; the second chunk holds 0.2 billion: True

```
 the Credit Facility for

2024.

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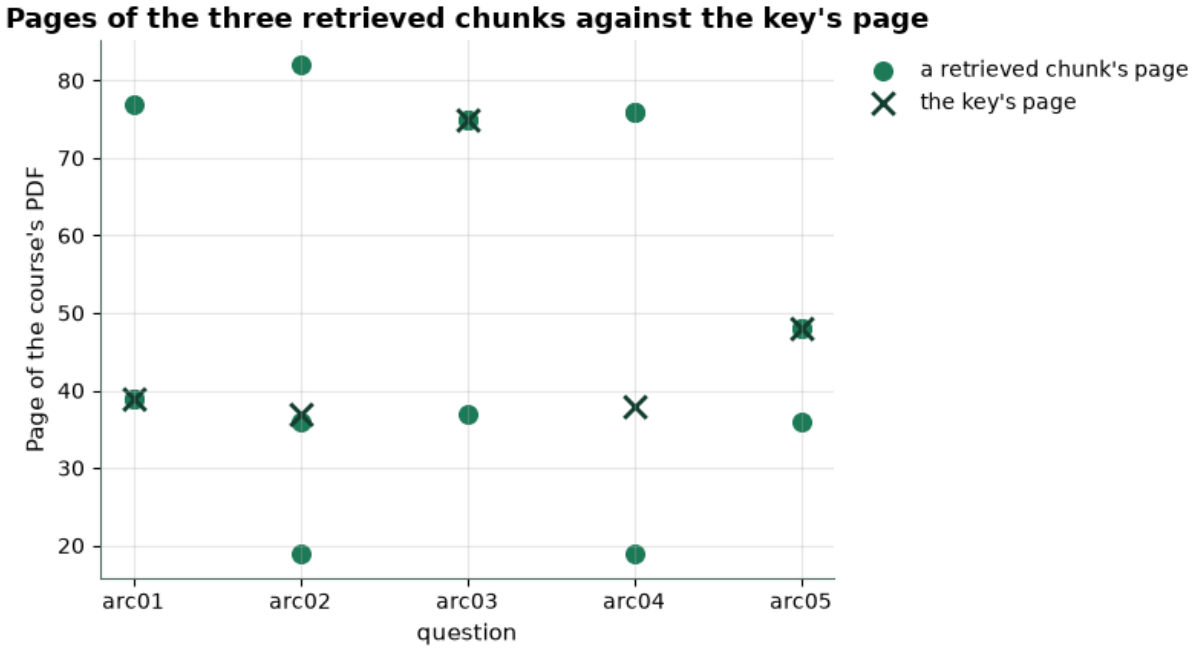
Indebtedness

The outstanding principal balances of long-term debt, before unamortized discounts and fees, and the changes ...
```

Source: Boyd Gaming Corporation, Form 10-K for fiscal 2025, filed 2026-02-20, accession 0001437749-26-004908, Item 7, page 36 of the course's PDF.

The model is about to see both figures, 160.7 and 0.2 billion. The key is $160.7 million.

Three rules, three counts



3 of 5

by page

4 of 5

by quote

4 of 5

by figure

id	key page	pages	page quote	figure	
arc01	39	39, 39, 77	1	1	1
arc02	37	36, 19, 82	0	0	1
arc03	75	75, 75, 37	1	1	1
arc04	38	76, 19, 76	0	1	0
arc05	48	48, 48, 36	1	1	1

Settled today: a retrieval miss is counted by the figure rule; day 11's recall stays by page; every count names its rule.

The prompt: three messages

system	Course 7's SYSTEM_PROMPT: exact quotes, section numbers, NOT IN TEXT, the refusal line
system	CITATION_RULE: labelled chunks; answer only from them; a label after every sentence; NOT IN CORPUS
user	Chunk 1: [label] text / Chunk 2 / Chunk 3 / Question: ...

3

messages: system, system, user

2 + 1

two system, one user

[509, 444, 1677]

characters, arc02

Nothing else goes in: no key quote, no page outside the chunks, no hint about the answer.

One answer: arc02 in the saved run

answer(client, question, retrieve, settings, mode); key: $160.7 million

```
Approximately $0.2 billion was outstanding under the Credit Facility as of December 31, 2025 [boyd10k; Item 1A; p. 19; c0225].
```

1	no match	no	yes
label cited; sent: yes	grade against the key	states 160.7	states 0.2 billion

Model gpt-4.1-mini-2025-04-14, temperature 0, seed 42; 1 request, 700 prompt and 41 completion tokens; saved run recorded 2026-10-13; replayed (no key loaded).

```
chunker=recursive 512/64, embedder=minilm@1110a24, index=faiss exact, retrieve=dense k=3, rerank=none, expand=0, model=gpt-4.1-mini, temperature=0, top_p=1, max_tokens=300
```

Grounded is not right: every word is in the text sent, the citation is real, and the key is missed.

The request's settings, and the names a desk meets

The request as sent, read from the transport's record: model gpt-4.1-mini, three messages

0.0	42	300	no
temperature	seed (ragkit.SEED)	max_tokens	top_p in the body

The same pipeline under LangChain's names: no framework imported, seven functions

`PyMuPDFLoader / PyPDFLoader`	pdf_pages (day 2); page_items adds the 10-K item
`RecursiveCharacterTextSplitter(chunk_size, chunk_...`	recursive_split and chunk_table (day 3): 512 and 64
`HuggingFaceEmbeddings / SentenceTransformerEmbedd...`	load_embedder and embed_chunks (day 4)
`Chroma.from_documents / FAISS.from_documents`	the chunk table, the matrix, numpy_index or faiss_index, and index_manifest...
`vector_store.similarity_search(question, k)`	search_numpy or search_faiss (day 4)
`vector_store.as_retriever(search_kwargs={'k': 3})`	retriever(chunks, matrix, model, settings) with k = 3 (today)
`RetrievalQA.from_chain_type(chain_type='stuff')`	rag_prompt and answer (today): the prompt is written out, not kept in a tem...

The local route, on a T4

LOCAL_MODEL, read on the Hub 2026-10-12

file	`Mistral-7B-Instruct-v0.3-Q4_K_M.gguf`
size	`4.37 GB (the repository's file list)`
licence	`apache-2.0 (the card's metadata)`
runner	`llama-cpp-python 0.3.36, MIT`
wheels	`the project's CUDA index, cu118 to cu132`
on a CPU	`22.4 s for one answer, one thread`

What the saved notebook prints

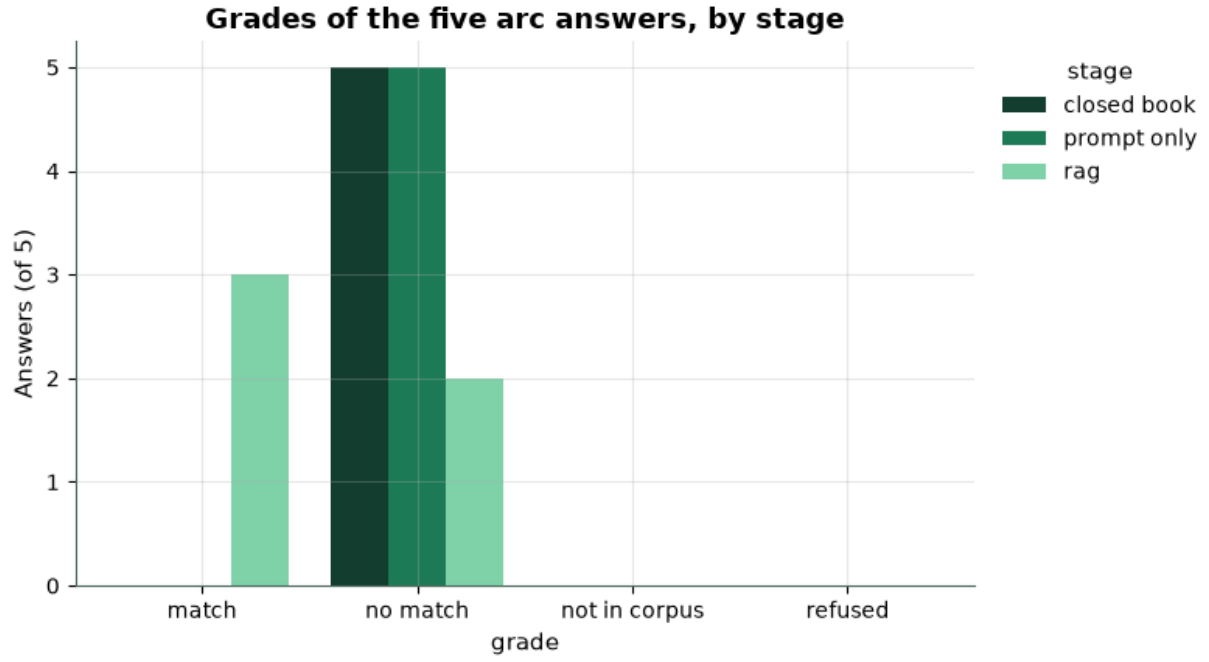
```
No GPU in this runtime: this cell is skipped. In
Colab, change the runtime to the T4 GPU and run the
notebook from the top.
```

```
GPU found: False. Local-model cells: skipped.
```

In Colab on a T4 the cell installs the wheel for the runtime's CUDA, downloads the file once, and answers arc02 from the same three chunks.

GPU advised in Colab, never required; the recorded lesson runs on the API, inside the four-minute budget.

Fifteen answers, graded against their keys



0 of 5

closed book (day 1)

0 of 5

prompt only (day 1)

3 of 5

RAG (today)

Citations

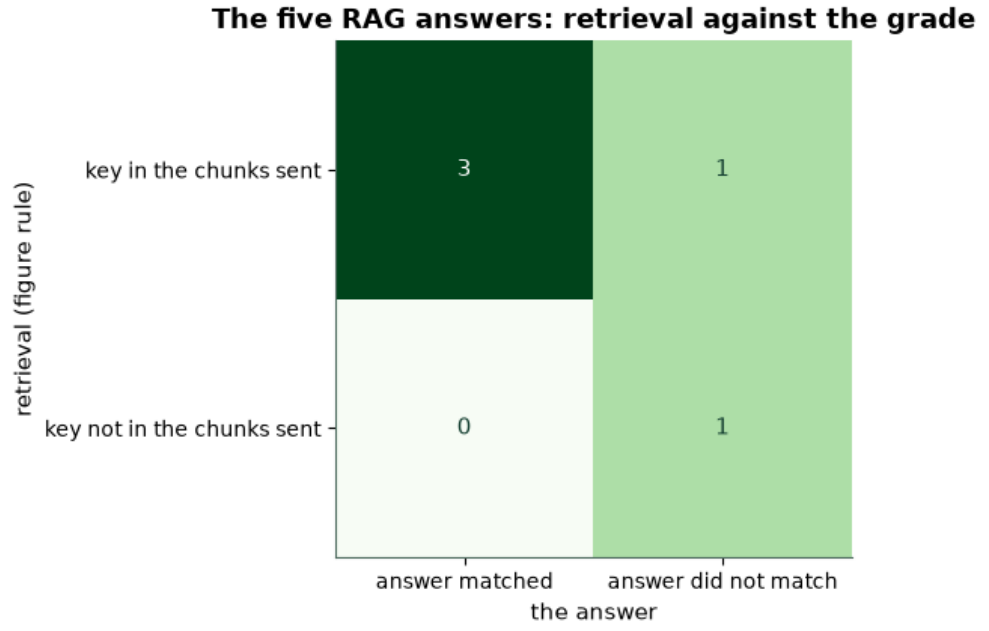
5 of 5 cite only chunks that were sent

3 of 5 end every sentence with a label

Two answers quote two sentences and cite once at the end: day 10's check will reject the uncited one.

Closed book 0, prompt only 0, RAG 3 of 5: retrieval is where the chart changes.

Retrieval against the grade



Two kinds of miss, by the figure rule

retrieval miss: arc04

The leverage ratio sat in a chunk that was not retrieved: no answer could have matched.

generation miss: arc02

160.7 in the top chunk, 0.2 billion in the second; the model took the second and cited it.

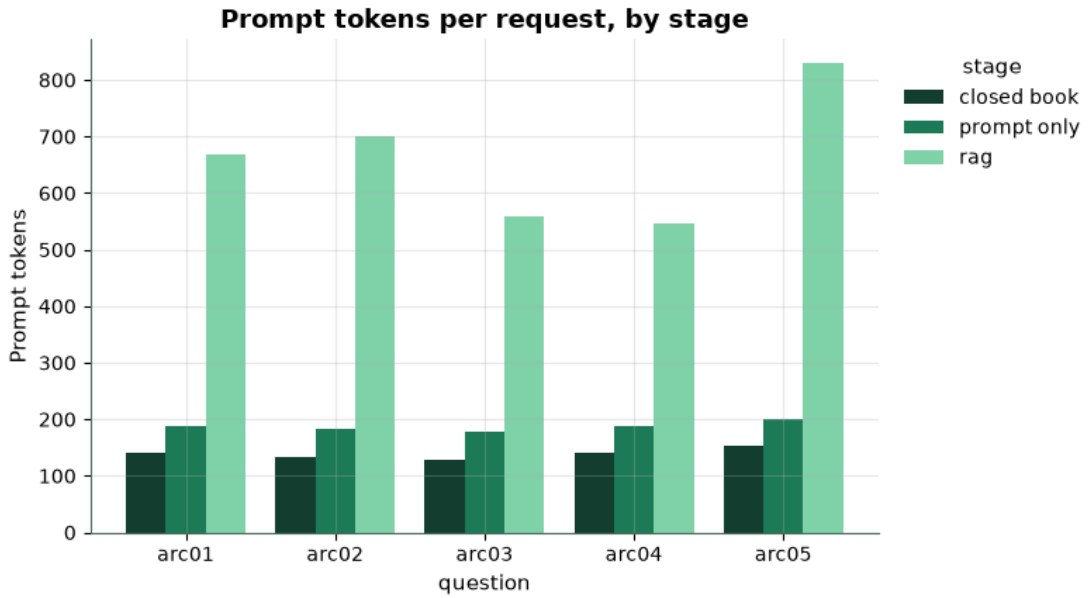
The same five, by two rules

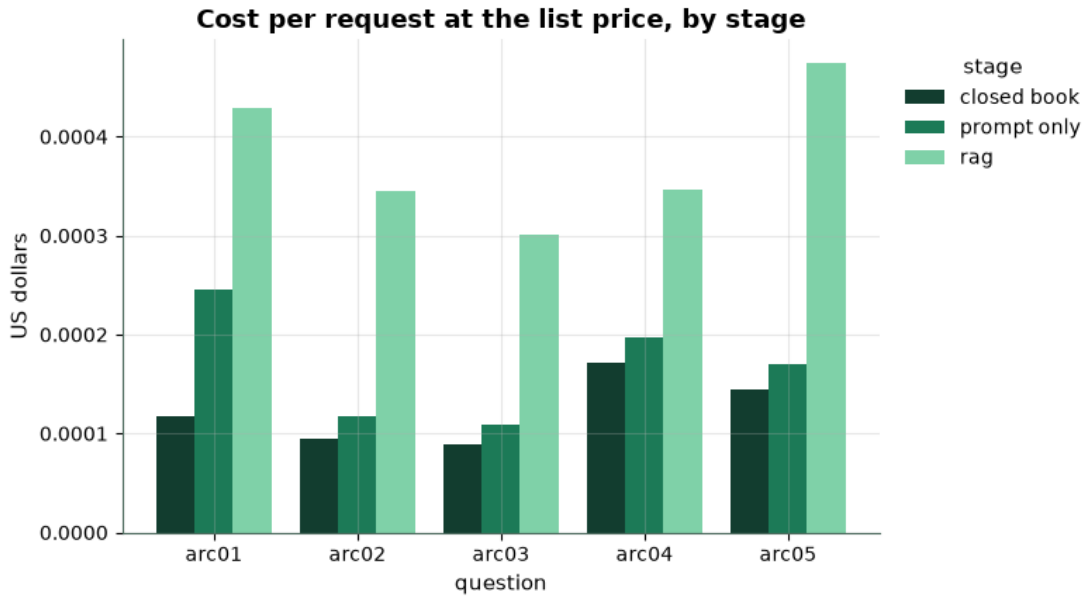
by page 3 of 5 retrieved; by figure 4 of 5

every citation names a chunk that was sent: 5 of 5

Rows: the key's figures in the chunks sent. Columns: the answer matched. Read the off-diagonal cells.

Tokens and cost by stage





stage: prompt tokens, completion tokens, USD, over five requests

```
closed book     697   211 0.000616
prompt only     937   290 0.000839
rag            3305   359 0.001896
```

Fifteen requests: 4,939 prompt and 860 completion tokens, US$0.003352; a RAG request carries 4.7 times the prompt tokens.

What the pipeline shows, and where it fails

Finding 1, as counts

closed book 0 of 5, prompt only 0 of 5, retrieval 3 of 5 matched their keys, on these five questions and this document

Finding 2, as counts

of 2 RAG misses, 1 came from retrieval (the key's figures were not in the chunks sent) and 1 from generation (they were sent and the answer did not match)

3 / 4 / 4

of 5 retrieved, by page / quote / figure

one key

match = the key's figures in the answer

one document

as of one filing date

Excel side by side, and Git

Excel side by side (tested in Excel)

`=COUNTIFS($B$2:$B$16,J2,$E$2:$E$16,"match")`	0, 0, 3 by stage = groupby
`=COUNTIFS(E2:E6,1,G2:G6,1)`	3: key in the chunks and matched
`=COUNTIFS(E2:E6,1,G2:G6,0)`	1: key in the chunks, not matched
`=COUNTIFS(E2:E6,0,G2:G6,0)`	1: key not in the chunks, not matched
`=COUNTIF(G2:G6,1)/COUNTA(G2:G6)`	0.6000 matched share
`dtype=str; the question and pages cells`	15 of 15 ids read back as text

```
run,settings_hash,questions,recall_at_k,answered,matched_key,grounded,mean_prompt_tokens,usd
day05_rag_k3,6579143b53aa799442ea4d46869c3821681c5811d4671c546cba697f10cd0b09,5,0.6000,1.0000,0.6000,1.0000,661.0000,0.0019
```

Git: rag_results.csv, one row named by the settings hash; nine files committed; tag week1; 32 tests.

The AI check

The docstring is the prompt. The retrieval code that comes back runs and looks finished.

1

Docstring

Units, conventions, and what the code must never let through: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Plant the case the docstring promises to catch. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the chunks go after the question with no labels, so an answer has nothing to cite.

Code that runs is not the same as code that is right.

What you can do now

- Every setting in one frozen record, printed as a settings line
- A retriever; hits counted by page, by quote, and by figure
- The prompt: system, citation rule, labelled chunks, question
- One answer, read with its LLM line and its settings line
- A retrieval miss told from a generation miss

Week 2, Day 1 BM25 and hybrid search